CHAPTER → 4

STRING

1. Introduction to Strings

A **string** is a fundamental data type in Python that represents a sequence of characters. Strings are used to store textual data such as names, messages, file paths, or any alphanumeric content. They are **immutable**, meaning once a string is created, it cannot be changed in place.

2. Creating Strings in Python

Strings in Python can be created using:

- **Single quotes (')**
- **Double quotes (")**
- **Triple quotes (''' or """)** — often used for multiline t

Example – > s1 = 'Hello'

s2 = "Python"

s3 = '''This is a multiline string'''

note : Strings can also be created from other data types using the str() constructor:

s4 = str(123) # '123'

3. Accessing Characters in a String

Strings are **indexed** starting from **0**.[0,1,2,3,4,5,6,7]

Code : name = "Himanshu"

print(name[0]) # Output: H

print(name[-1]) # Output: u (last character)

4. String Slicing

Syntax : string[start:end:step]

Code: name = "Himanshu"

print(name[0:4]) # 'Hima'

print(name[::2])   # 'Hmnsh'

print(name[::-1]) # 'uhsnamiH' (Reversed)

5. String Operations

Python supports several operations on strings:

1. **Concatenation :**   Joining two or more strings:

   Code : first = "Hello"

   second = "Himanshu"

   print(first + " " + second) # 'Hello Himanshu'

2. **Repetition:**   Repeat a string multiple time:

   Code: print ("Hi! " * 3) # 'Hi! Hi! Hi! '

3. **Membership :** Check if a substring exists:

   code : ["man" in "Himanshu"   # True] ["x" not in "Himanshu"   # True]

6. String Methods :

Python provides a rich set of built-in string methods:

Method / Function	Description	Example Code
`replace(old, new)`	Replace substring `old` with `new`	`print("Hi Himanshu".replace("Hi", "Hello"))` # Hello Himanshu
`find(sub)`	Returns first index of substring `sub`, or -1 if not found	`print("Himanshu".find("man"))` # 3
`capitalize()`	Capitalizes first character of the string	`print("himanshu".capitalize())` # Himanshu
`count(sub)`	Counts occurrences of substring `sub`	`print("banana".count("a"))` # 3
`endswith(suffix)`	Checks if string ends with `suffix`	`print("filename.txt".endswith(".txt"))` # True
`len(s)`	Returns length of string `s`	`print(len("Himanshu"))` # 8
`ord(c)`	ASCII/Unicode value of character `c`	`print(ord('H'))` # 72
`chr(i)`	Character corresponding to ASCII `i`	`print(chr(72))` # 'H'

sorted(s)	Returns sorted list of characters in string	`print(sorted("bca"))` # ['a', 'b', 'c']
upper()	Converts to uppercase	`print("hello".upper())` # HELLO

```
Code: # 1. upper()

text = "hello"

print(text.upper())  # HELLO

# 2. lower()

text = "HELLO"

print(text.lower())  # hello

# 3. title()

text = "my name is himanshu"

print(text.title())  # My Name Is Himanshu

# 4. strip()

text = "  hello  "

print(text.strip())  # hello

# 5. replace()

text = "Hi Himanshu"

print(text.replace("Hi", "Hello"))  # Hello Himanshu

# 6. find()

text = "Himanshu"

print(text.find("man"))  # 3

# 7. split()

data = "apple,banana,mango"

print(data.split(","))  # ['apple', 'banana', 'mango']

 8. join()

words = ["Hello", "Himanshu"]

print(" ".join(words))  # Hello Himanshu
```


7. String Formatting Techniques :

1 . Using format() Method

code : print ("My name is {} and I am {} yearsold.".format("Himanshu", 22))

2. Using f-strings :

Code : age = 22

print(f"My name is Himanshu and I am {age} years old.")

10. Useful Built-in Functions

- len(s): Returns the number of characters in the string
- ord(c): Returns ASCII value of a character
- chr(i): Returns character from ASCII value
- sorted(s): Returns a list of sorted characters

code : len("Himanshu") # 8

ord('H')   # 72

chr(72)   # 'H'

sorted("bca")   # ['a', 'b', 'c']

11. Escape Sequences :

Escape sequences let you insert special characters into strings.

Sequence	Meaning
\n	Newline
\t	Tab
****	Backslash
\'	Single quote
\"	Double quote

Practice Set:

Que 1: **Create Strings**

Create three strings using single quotes, double quotes, and triple quotes (multiline). Print all three.

Que: 2 : Indexing and Slicing

Given name = "Himanshu", write code to:

a) Print the 3rd character.

b) Print characters from index 2 to 5.

c) Print the string reversed.

Que: 3 : String Operations

Concatenate the strings "Hello" and "Himanshu" with a space in between and print. Also, print the string "Hi! " repeated 4 times.

Que: 4 Membership Check

Check if "man" is present in the string "Himanshu" and print the result (True/False). Check if "xyz" is not in the string "Himanshu" and print the result.

Que: 5 **String Methods**

Using the string "banana", print:

a) The count of "a" in it.

b) The string after replacing "na" with "xy".

c) The index where "na" first appears.

Que: 6 Formatting Strings

Print your name and age using both.format() and f-string formatting methods.

Que: 7 Escape Sequences

Write a string using escape sequences to print this exactly: